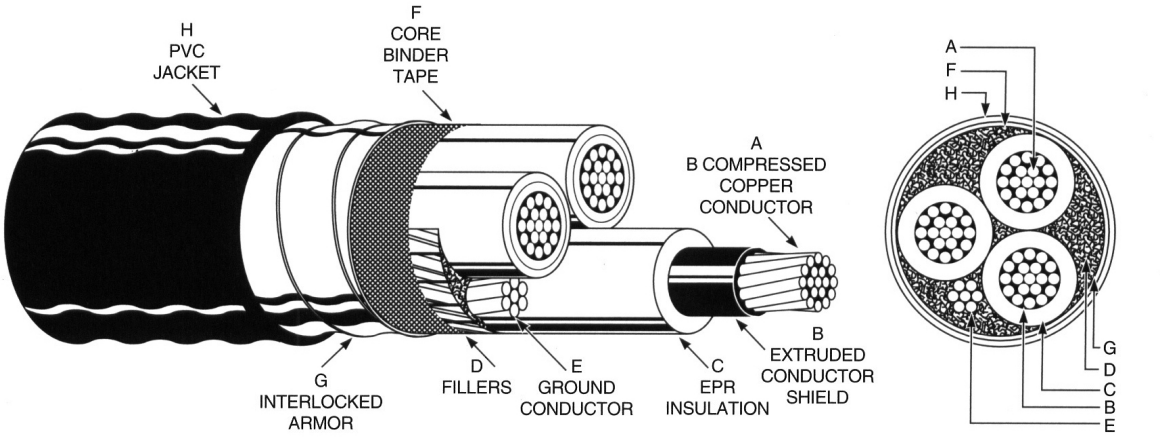
AL09ENS

2.4kV. Type MV-105 or MC. Copper Conductors. Thermosetting Conductor Shield. EPR Insulation (133%). Non Shielded. Copper Ground Wire. Aluminum Armor. PVC Jacket. Limited to 2,400 Volts Maximum.



APPLICATIONS

Southwire AL09ENS Type MV-105 or MC Cable is for use in aerial installations, direct buried, concrete-encased installations, cable trays, troughs or continuous rigid cable supports. These cables are capable of operating continuously at a maximum conductor temperature of 105°C for normal operation, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 2,400V in wet or dry locations. Maximum sidewall pressure is 1000 lbs.

SPECIFICATIONS

Southwire AL09ENS Type MV-105 or MC Cable is manufactured and tested in accordance with the latest revisions of:

- ICEA S-96-659 (NEMA WC 71) - Nonshielded Cables Rated 2001-5000 Volts for Use in the Distribution of Electric Energy.
- UL 1072 - Medium Voltage Power Cables.
- IEEE 1202 - Flame Test (70,000 Btu/hr Vertical Tray Test).
- ICEA T-29-520 - Vertical Cable Tray Flame Tests (210,000 Btu/hr.).
- RoHS

CONSTRUCTION

Southwire AL09ENS Type MV-105 or MC Cable is a flame-resistant product that offers Class B Compressed copper conductors surrounded buy an extruded conductor shield covered with flexible EPR insulation. Non-wicking fillers and a bare copper ground complete the assembly while covered with binder tape and a PVC outer jacket.. For use in cable trays per NEC Article 392. SOLONON TM low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

greenSpec™ RoHS Compliant

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AL09ENS 3c, EPR , N/S, GW, Alum. Interlocked Armor

Product Code	Size AWG or kcmil	Conductor Diameter		Insulation Diameter 115 mil (133%)		Copper Grounding Conductor Size AWG	Armor Diameter		Jacket Thickness		Approximate Overall Diameter		Approximate Net Weight		Ampacities+	
		inch*	mm	inch**	mm		inch	mm	inch	mm	inch	mm	lbs/Mft.	kg/km	Direct Burial	In Air
AL09ENS-006	6	.178	4.52	.415	10.5	6	1.114	28.3	.05	1.27	1.214	30.8	801	1192	105	79
AL09ENS-004	4	.225	5.72	.463	11.7	6	1.212	30.8	.05	1.27	1.312	33.3	1006	1497	135	105
AL09ENS-002	2	.283	7.19	.518	13.1	6	1.336	33.9	.05	1.27	1.436	36.5	1307	1945	180	140
AL09ENS-001	1	.322	8.18	.558	14.2	4	1.422	36.1	.05	1.27	1.522	38.7	1589	2364	200	160
AL09ENS-010	1/0	.362	9.19	.598	15.2	4	1.609	40.9	.06	1.52	1.729	43.9	1914	2848	230	185
AL09ENS-020	2/0	.405	10.29	.638	16.2	4	1.695	43.1	.06	1.52	1.815	46.1	2243	3337	260	215
AL09ENS-040	4/0	.512	13	.743	18.9	3	1.922	48.8	.06	1.52	2.042	51.9	3165	4709	335	285
AL09ENS-250	250	.558	14.17	.8	20.3	2	2.046	52	.06	1.52	2.166	55	3687	5486	365	320
AL09ENS-350	350	.661	16.79	.903	22.9	2	2.267	57.6	.075	1.91	2.417	61.4	4844	7208	440	395
AL09ENS-500	500	.79	20.07	1.028	26.1	1	2.537	64.4	.075	1.91	2.687	68.3	6526	9710	530	485

*Minimum diameter per ASTM standards. ** ±0.050.
+Ampacities are based on the NEC, 2011 Edition. Direct burial ampacities are based on Table 310.83 three conductors within an overall covering directly buried, 90°C conductor, 20°C earth ambient temperature. In air ampacities are based on Table 310.71 three conductors within an overall covering in free air, 90°C conductor, 40°C ambient temperature.

SCOPE: This specification covers three conductor EPR (ethylene propylene rubber) insulated, non-shielded, interlock armored, thermoplastic jacketed power cables with grounding conductor for use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs, or continuous rigid cable supports. These cables are capable of operating continuously at a temperature of 105°C for normal operations, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated 2,400 volts in wet or dry locations.

STANDARDS: The following standards will form part of this specification:UL Standard 1072 for Medium Voltage Power Cables and ICEA S-96-659 (NEMA WC 71) Nonshielded Cables Rated 2001-5000 Volts for Use in the Distrbution of Electric Energy.

CONDUCTOR: The conductor will be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8 and ICEA.

CONDUCTOR SHIELD: The conductor will be shielded with an extruded semi-conducting thermosetting polymeric layer, which will be firmly bonded to the insulation. The thickness will be in accordance with the referenced standards.

INSULATION: The insulation will be EPR (ethylene propylene rubber) meeting the requirements of the referenced standards. The average thickness will be 0.115" and the minimum spot thickness will be not less than 90% of the average thickness.

GROUNDING CONDUCTOR: The ground conductor will be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8.

ASSEMBLY: The insulated power conductors will be cabled round with fillers and with a grounding conductor in one outer interstice and covered with a binder tape.

ARMOR: A single strip of interlocked armor of aluminum will be applied over the assembly.

JACKET: The cable will be covered with a yellow PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness will be in accordance with the referenced standards and the minimum spot thickness will be not less than 80% of the average thickness. The jacket will be sunlight resistant and will meet the requirements of IEEE 1202 (70,000 BTU/hr.) and ICEA T-29-520 (210,000 Btu/hr.) vertical cable tray flame tests. Optional SOLONON low smoke, non halogen polyolefin jackets and CPE jackets available upon request.

INDENTIFICATION: Manufacturer's identification will be printed on the jacket.

TESTS: Physical and electrical tests will be conducted in accordance with the requirements of the referenced standards.

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